Belief State Geometry in Non-Ergodic Transformers

Simplex Take-Home Experiment

Juee Jahagirdar

March 5, 2026

Abstract

We train a small transformer on sequences from a non-ergodic Mess3 process and analyse what geometry emerges in the residual stream. The training corpus contains four sequence types formed by crossing two values of the transition parameter α with two ergodic components (clockwise and counter-clockwise cycling), where each sequence is generated entirely by one type. We find that the residual stream linearly encodes the analytical belief state of the generative process, with four clusters separating as context grows, consistent with pre-registered predictions. A generalisation experiment reveals that a model trained only on the clockwise component maps counter-clockwise sequences onto the same geometric manifold with nearly identical belief-state correlations, suggesting the non-ergodic boundary is transparent to the learned representations. Code and figures are available at https://github.com/torturedengineer/mess3-belief-geometry.

Contents

1 The Data-Generating Process **2**
- 1.1 Mess3 as a Hidden Markov Model . . . 2
- 1.2 Non-ergodicity . . . 2
- 1.3 Training Dataset . . . 2
- 1.4 Why This Structure is Interesting . . . 3

2 Pre-Registered Prediction **3**
- 2.1 Belief State Geometry . . . 3
- 2.2 Predicted Residual Stream Geometry . . . 3

3 Residual Stream Geometry **4**
- 3.1 Model and Training . . . 4
- 3.2 Analytical Ground Truth . . . 4
- 3.3 Separation Across Layers and Token Positions . . . 4
- 3.4 Alignment with Belief States . . . 5

4 Additional Analysis: Generalisation Across the Non-Ergodic Boundary **6**
- 4.1 Setup . . . 6
- 4.2 Results . . . 6
- 4.3 Interpretation . . . 7
- 4.4 Connection to CTW and CTS . . . 8
- 4.5 Belief Crystallization . . . 8

5 Discussion **8**

1 The Data-Generating Process

1.1 Mess3 as a Hidden Markov Model

Mess3 is a Hidden Markov Model with three hidden states $\{0, 1, 2\}$ and three observable symbols $\{0, 1, 2\}$, parameterised by $\alpha \in (0, 1)$ (transition determinism) and $x \in (0, 1)$ (emission noise). Figure 1 shows the structure of both ergodic components.

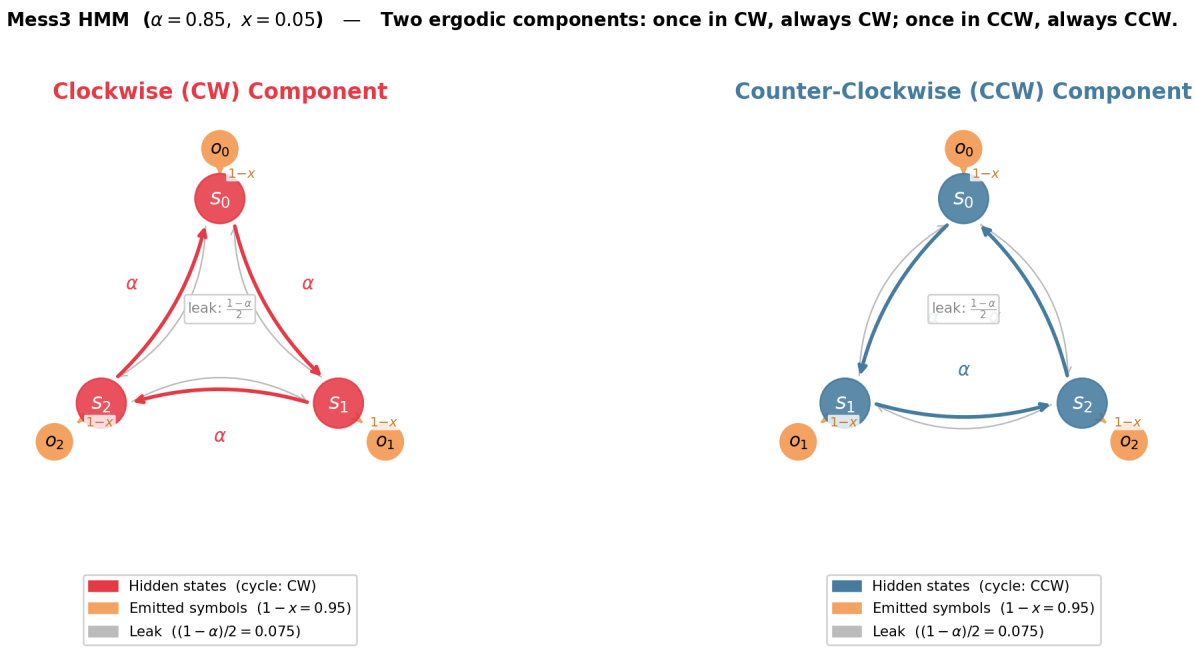

Figure 1: Mess3 HMM structure. Hidden states cycle through $\{0, 1, 2\}$; each state emits its own symbol with probability $1 - x$ and leaks to the others with probability $x/2$ each. The two panels show the clockwise (CW) and counter-clockwise (CCW) ergodic components.

Transitions. Each hidden state moves to a designated next state with probability α and leaks to the other two with probability $(1 - \alpha)/2$ each. The direction of cycling defines the two ergodic components:

$$\text{CW:} \quad 0 \to 1 \to 2 \to 0 \qquad \text{CCW:} \quad 0 \to 2 \to 1 \to 0$$

Emissions. $E_{ii} = 1 - x$ and $E_{ij} = x/2$ for $i \neq j$. With $x = 0.05$ each state emits its own symbol about 95% of the time.

1.2 Non-ergodicity

Mess3 is non-ergodic: a sequence in the CW cycle stays there permanently and can never reach CCW. These are *ergodic components*, absorbing irreducible sub-chains. The steady-state distribution within each is uniform, $\pi = (1/3, 1/3, 1/3)^\top$.

1.3 Training Dataset

We construct four sequence types by crossing two values of α with the two components, holding $x = 0.05$ fixed:

Label	Name	α	Component
0	Det-CW	0.85	Clockwise
1	Det-CCW	0.85	Counter-clockwise
2	Rnd-CW	0.50	Clockwise
3	Rnd-CCW	0.50	Counter-clockwise

Table 1: Four process types. Every training sequence is generated entirely by one type, respecting the non-ergodic constraint.

500 sequences per type (2000 total), length 16. 80/20 train/test split.

1.4 Why This Structure is Interesting

To predict the next token well, the model must infer which of the four processes it is currently reading from context alone, since the component label is never observed. This is the point Computational Mechanics makes about synchronisation: the minimal sufficient statistic for prediction is richer than the generative process itself shai2024transformers. A well-trained predictor should maintain more structure than the HMM that produced the data.

2 Pre-Registered Prediction

2.1 Belief State Geometry

Given observations $x_1, \ldots, x_t$, the belief state is the posterior over the current hidden state: $b_t(s) = P(h_t = s \mid x_1, \ldots, x_t)$. It updates via the HMM forward recursion rabiner1989tutorial:

$$b_t \;\propto\; E_{:,x_t} \odot (T^\top b_{t-1}). \tag{1}$$

Since $b_t \in \Delta^2$, the set of reachable belief states traces a triangle. The size depends on α: high α concentrates beliefs near vertices (large triangle); low α keeps beliefs near the centroid (small triangle).

2.2 Predicted Residual Stream Geometry

Pre-registered prediction (honour code):

1. PCA of the residual stream should reveal **four clusters**, one per process type.
2. At early token positions the clusters should **overlap**; the model has not seen enough context to distinguish components.
3. Clusters should **separate as context grows**, with CW/CCW resolving first (determinable after roughly 2 tokens) and α separation more gradually.
4. Deeper layers should show cleaner separation.
5. PC axes should **correlate with belief state dimensions** $P(h = 0)$, $P(h = 1)$, $P(h = 2)$.

I initially thought the combined geometry might form one large simplex, but revised this: since no sequence crosses the ergodic boundary, the model has no pressure to merge the two components. Separate triangles, and with two α values, four triangles seemed more likely.

3 Residual Stream Geometry

3.1 Model and Training

A 3-layer decoder-only transformer with causal attention, $d_{\text{model}} = 64$, 4 attention heads, feed-forward width 128, context window 16. Trained with Adam ($\text{lr} = 3 \times 10^{-3}$, cosine annealing, 30 epochs). Final loss 0.883, below the uniform-random baseline $\ln 3 \approx 1.099$ (Figure 2).

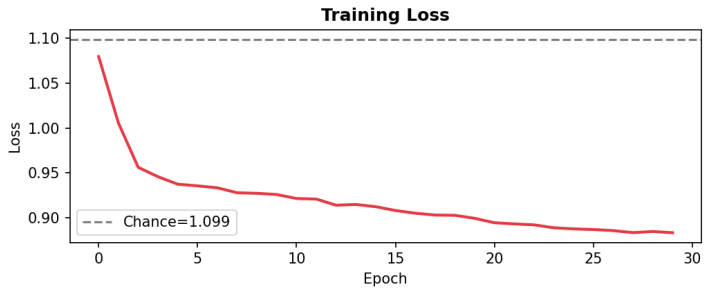

Figure 2: Training loss converges well below the chance baseline.

3.2 Analytical Ground Truth

We run the forward algorithm (Eq. 1) on every test sequence to compute ground-truth belief states analytically. Figure 3 shows these projected to 2D via PCA.

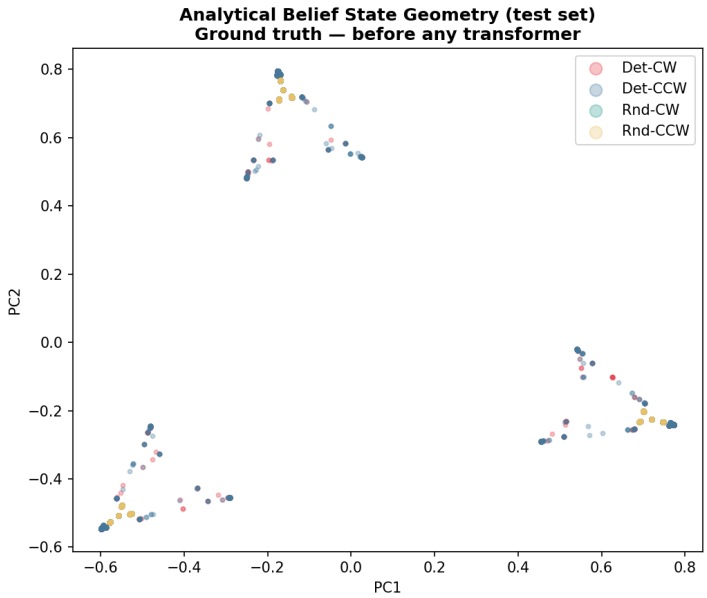

Figure 3: Analytical belief state geometry. Each point is $b_t \in \Delta^2$. The triangular cluster structure reflects the 3-state cycling of Mess3.

3.3 Separation Across Layers and Token Positions

Figure 4 shows PCA of the residual stream at each layer with all token positions pooled; Figure 5 isolates the last layer at individual positions.

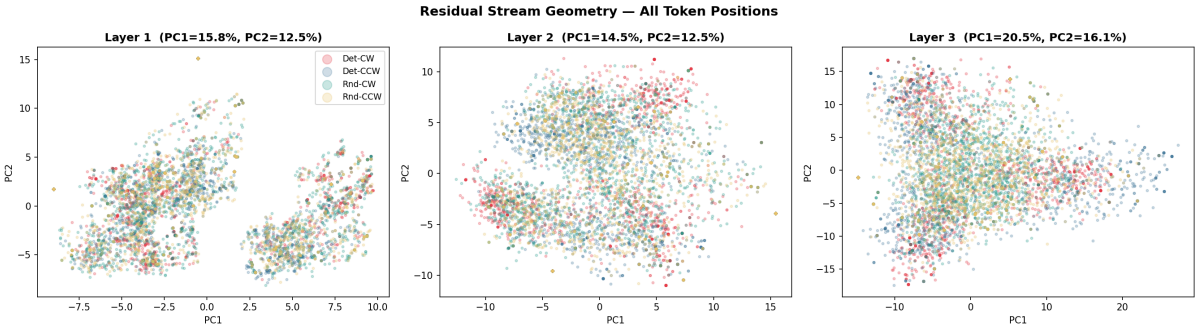

Figure 4: Residual stream geometry by layer (all token positions pooled). Layer 1 already shows two broad blobs. Apparent smearing is expected since early and late positions are mixed together.

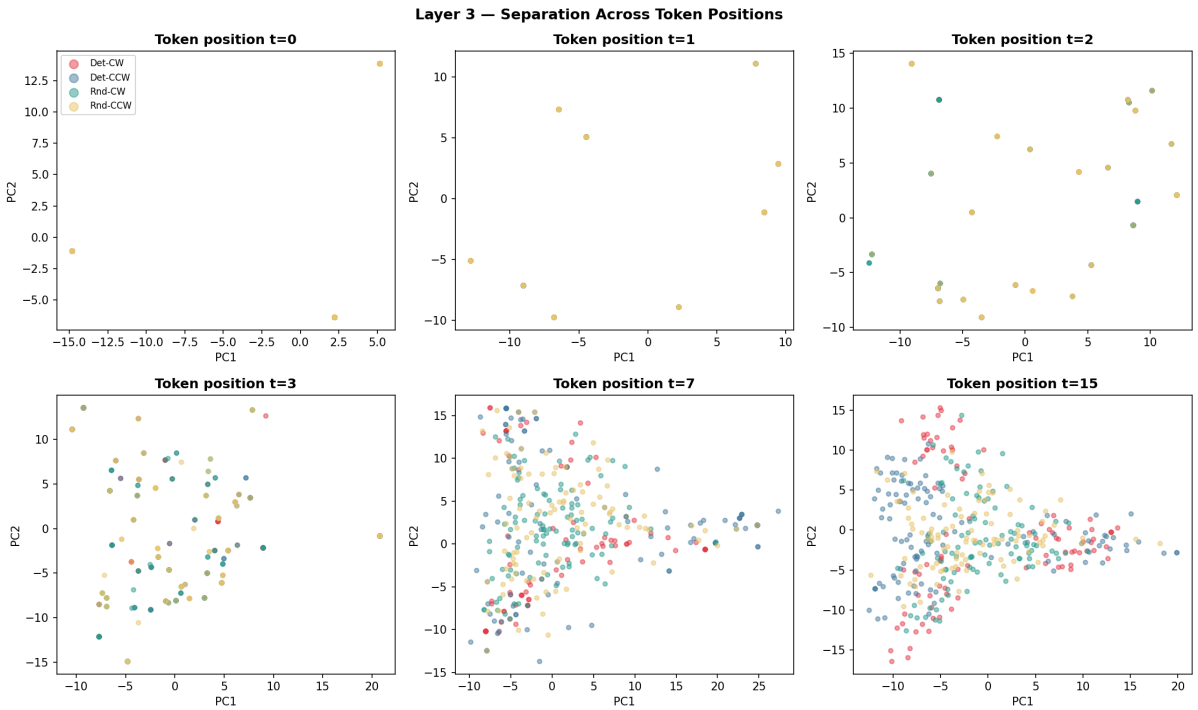

Figure 5: Layer 3 residual stream at specific token positions. At $t = 0$ essentially one cluster. By $t = 3$ all four colours appear. By $t = 7$ and $t = 15$ the clusters are clearly separated. CW/CCW separation precedes α separation, consistent with the pre-registered prediction.

3.4 Alignment with Belief States

We project layer-3 activations to 3D and measure absolute correlations with analytical belief state dimensions (Figure 6). PC1 correlates 0.54 with $P(h = 0)$; PC2 tracks $P(h = 1)$ and $P(h = 2)$.

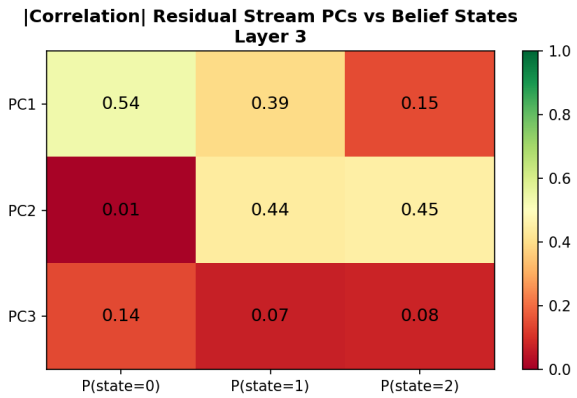

Figure 6: Absolute correlation between residual stream PCs and belief state dimensions. The residual stream linearly encodes the belief state.

Result: The transformer's residual stream linearly represents the belief state geometry of the component processes, consistent with shai2024transformers in the ergodic setting. The non-ergodic structure introduces an additional axis of variation (component identity) that is also linearly represented.

4 Additional Analysis: Generalisation Across the Non-Ergodic Boundary

4.1 Setup

We train a fresh model on *only* Det-CW ($\alpha = 0.85$, CW) and feed it Det-CCW sequences it has never seen. Three possible outcomes: (1) random scatter, meaning the model memorised surface statistics; (2) partial structure, meaning partial generalisation; (3) CCW activations fall in the same region as CW with similar internal structure, meaning the model learned abstract Mess3 structure independent of direction.

4.2 Results

Figures 7 and 8 show the outcome.

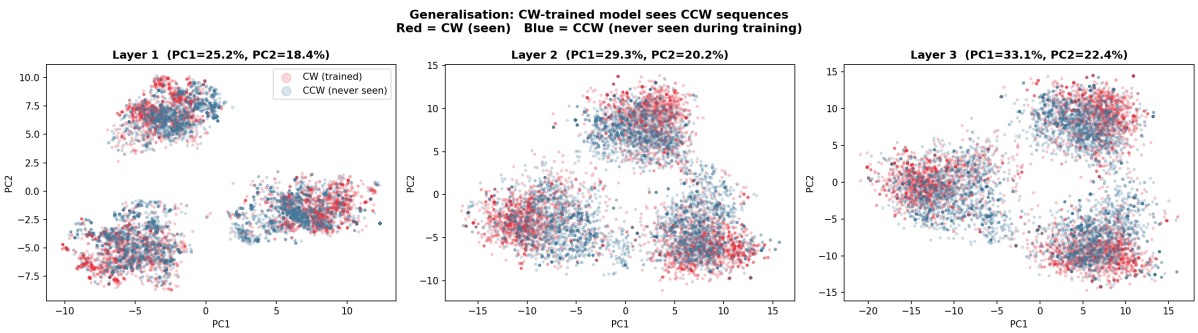

Figure 7: CW-trained model applied to unseen CCW sequences. Red = CW (trained). Blue = CCW (never seen). CCW activations fall almost exactly on top of CW across all layers, outcome 3.

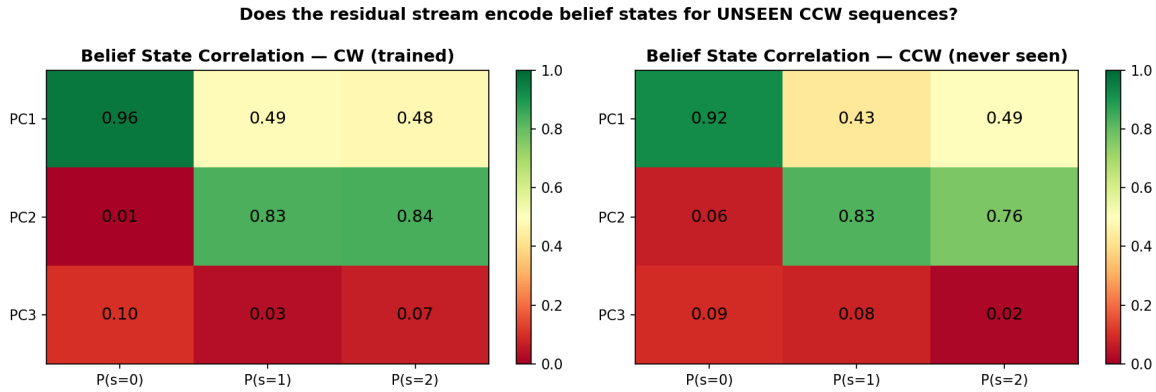

Figure 8: Belief state correlations for CW (trained) vs. CCW (never seen). PC1 vs. $P(s{=}0)$: 0.96 (CW) and 0.92 (CCW). PC2 vs. $P(s{=}1)/P(s{=}2)$: 0.83/0.84 vs. 0.83/0.76. Belief states for unseen CCW sequences are encoded with nearly the same fidelity as trained CW sequences.

The inter/intra-cluster distance ratio is **1.005**, confirming CW and CCW are completely overlapping in the residual stream. A permutation test on the linear probe at Layer 3, $t = 15$ gives real accuracy 0.863 against a null mean of 0.503 ($p = 0.0000$ over 100 permutations), confirming the within-group separation is real while the between-group boundary is not represented (Figure 9).

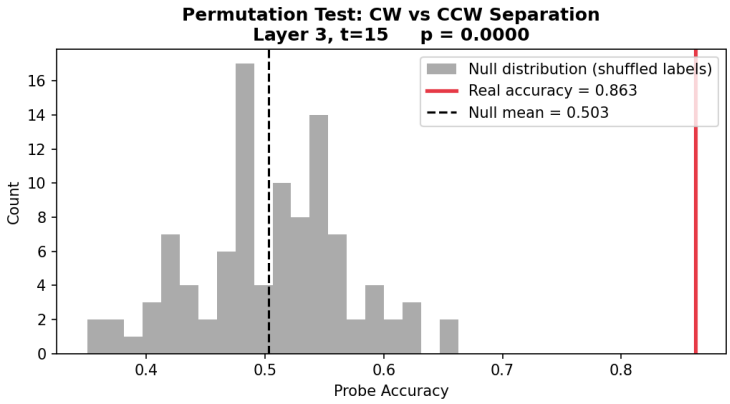

Figure 9: Permutation test for CW vs. CCW separation (Layer 3, $t = 15$). Real probe accuracy 0.863 (red) lies far outside the null distribution (mean 0.503, grey). $p = 0.0000$ across 100 permutations.

4.3 Interpretation

The non-ergodic boundary is invisible to the transformer's learned representations. The model correctly tracks belief states over the three hidden states and respects the emission structure, but does not represent the direction of cycling. Direction is precisely the information that distinguishes the two ergodic components, and a model trained on one component has no gradient signal to learn it.

Ruling out a confound. With $x = 0.05$, the emission statistics of CW and CCW are nearly identical at short context, so one might worry the overlap is trivial rather than informative. However, the full model trained on all four process types separates CW from CCW by $t = 3$, reaching probe accuracy of 88% by Layer 3. The two components are distinguishable from their sequences. The CW-only model's failure to separate them is therefore not a consequence of identical inputs, but of never receiving gradient signal to represent the boundary.

4.4 Connection to CTW and CTS

This connects to prior work I did implementing Context Tree Weighting (CTW) and Context Tree Switching (CTS). Figure 10 shows results from that work on a non-stationary environment where action dynamics flipped at $t = 500$.

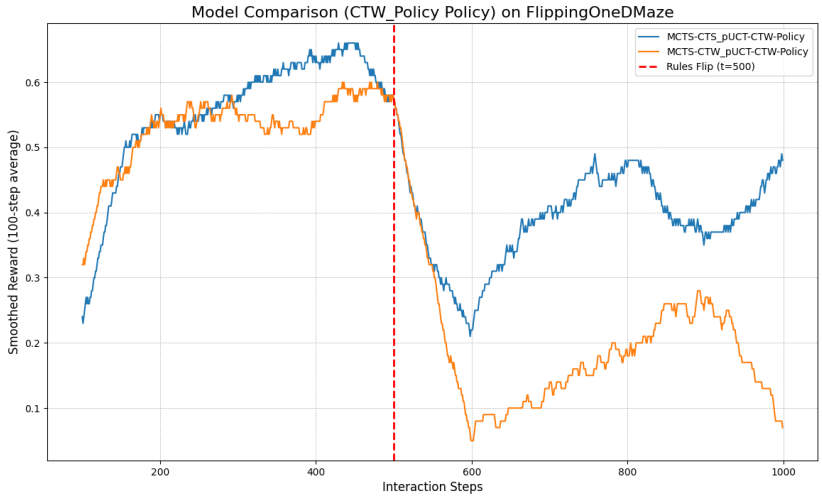

Figure 10: CTW vs. CTS on a flipping 1D maze (prior work). After the rule change at $t = 500$, CTS (blue) recovers within roughly 200 steps. CTW (orange) never recovers: it commits too heavily to accumulated history and cannot adapt.

CTW maintains a Bayesian mixture over context-tree models, weighting by predictive performance. CTS allows switching to a new model when evidence supports it, which is structurally the same as belief crystallization: the belief state commits to a vertex at precisely the point where CTS would switch its active model.

A transformer trained only on CW behaves like CTW with one model: no mechanism to represent the boundary it never crossed. The flipping maze result makes the consequence concrete: without the ability to switch, performance collapses and never recovers. The transformer avoids this only when trained on both components.

4.5 Belief Crystallization

We visualise the belief state trajectory on the 2D simplex as tokens arrive, alongside the attention pattern, for one CW and one CCW sequence side by side. The moment the two trajectories diverge on the simplex is when the model commits to a component. This is included as a supplementary animation (`belief_crystallization.gif`).

5 Discussion

What the geometry shows. The residual stream encodes two superimposed structures: the belief simplex for the current hidden state (a triangle rotating as the cycle progresses), and cluster identity separating the four process types. Both are linearly readable.

What was predicted correctly. Four-cluster separation with context was confirmed. PC alignment with belief state dimensions was confirmed. CW/CCW separating faster than α was confirmed.

What was not perfectly predicted. Clean triangular sub-structure within each cluster was not visually obvious. This is probably because PCA conflates the two geometric structures, and the within-cluster triangle is small relative to the between-cluster spread.

The generalisation finding. The most surprising result: a model trained only on CW maps CCW sequences onto the same geometric manifold with nearly identical belief state correlations. The non-ergodic boundary is transparent to the learned representations. This suggests transformers learn the shared generative structure (emissions, 3-cycle) without necessarily learning the directional structure that defines the ergodic boundary. Whether this is a limitation or a form of appropriate generalisation depends on the downstream task.

Acknowledgements

Code scaffolding and report polishing were assisted by Claude (Anthropic). All experiment design, predictions, result interpretation, the generalisation experiment, and connections to prior work (CTW/CTS) were the author's own. The honour-code prediction in Section 2 was made before running any experiments.

References

[Shai et al.(2024)] Shai, A. et al. (2024). *Transformers Represent Belief State Geometry in their Residual Stream*. LessWrong. https://www.lesswrong.com/posts/gTZ2SxesbHckJ3CkF

[Rabiner(1989)] Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286.